SUB-MODULE 7.3 · INSTRUCTION TUNING

Instruction-Following Datasets

The (instruction, input, output) triple — formats, major public datasets, and how choice shapes behaviour

9 min	B3 – Apply	L2 – Intermediate	AI-FT-IN-TH-000011
Duration	Bloom's	Level	ID

Instruction Dataset Anatomy

The Basic Unit: (Instruction, Input, Output) Triple

Each training example consists of: (1) Instruction — the task description ('Translate the following to French'), (2) Input — optional context or data, (3) Output — the desired model response. The model learns to follow the instruction given any input.

- Format choice shapes model behaviour: a model trained on Alpaca format behaves differently from one trained on ShareGPT (multi-turn) even on the same base model
- Multi-turn datasets (ShareGPT, OASST) teach conversational ability — single-turn datasets (Alpaca) teach task following
- System prompts in ChatML format allow persona / safety instruction injection at inference time

Three Major Dataset Formats — Side by Side

```
# ── Alpaca Format ────────────────────────────────────────────────────────────
{
  "instruction": "Summarise the following article in 3 bullet points.",
  "input": "Transformers have revolutionised NLP...",
  "output": "- Transformers use self-attention...\n- BERT introduced bidirectional...\n- GPT-3 showed few-shot..."
}

# ── ChatML Format (multi-turn) ───────────────────────────────────────────────
<|im_start|>system
You are a helpful AI assistant.
<|im_end|>
<|im_start|>user
What is the capital of France?
<|im_end|>
<|im_start|>assistant
Paris.
<|im_end|>
```

Rule: use ChatML or ShareGPT format for production fine-tunes — multi-turn capability is required for assistant behaviour. Alpaca is best for quick single-task fine-tunes.

```
}
```

Major Public Instruction Datasets — Compared

Dataset	Source	Size	Format	Best For
FLAN	Google (2021)	1M+ (60+ tasks)	Alpaca-like	General instruction following
Alpaca	Stanford (2023)	52K	Alpaca	Lightweight instruction tuning
ShareGPT	Community (2023)	~90K convs	ShareGPT	Multi-turn chat capability
OASST	Open Assistant	161K messages	Tree structure	Human-quality multi-turn
Dolly	Databricks (2023)	15K	Alpaca	Commercial use (Apache 2.0)
WizardLM	Microsoft (2023)	196K	Alpaca	Complex reasoning tasks

KEY REFERENCES

- Wei, J. et al. (2021). FLAN: Finetuned Language Models Are Zero-Shot Learners. ICLR 2022.
- Taori, R. et al. (2023). Stanford Alpaca: An Instruction-following LLaMA model. github.com/tatsu-lab/stanford_alpaca
- Köpf, A. et al. (2023). OpenAssistant Conversations. NeurIPS 2023.

▶️ **Next: Concept 15: Creating Quality Training Data (AI-FT-IN-TH-000012)**